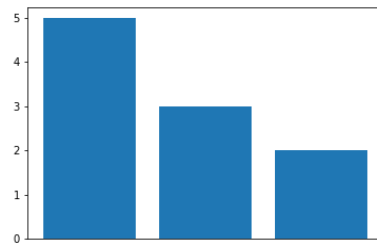


Gedächtnisprotokoll - Klausur Data Literacy WS19/201

11 Fragen, 100 Punkte, 90 Minuten

1. Name four ways to deal with missing data.
 - b) Explain one of them in detail
2. Doctors conduct a trial for a new drug against high blood pressure. n Patients $x_i, i \in \{1, \dots, n\}$ with high blood pressure are selected. $t_i \in \{0, 1\}$ denotes if Patient i is given the real drug or a placebo. Δ_i denotes the difference in blood pressure.
 - a) What are dependent and independent Variables in the trial?
 - b) Where could *selection bias* occur in this trial?
3. You have 12 Balls, one is heavier or lighter. You also have a scale that shows either: left is lighter, right is lighter, both sides are the same weight. What is the minimum number of weightings you would need to conduct to answer which one is the odd ball. Explain your result.
4. Consider the following:



- a) Which Visual Variables are used?
 - a) What are the best and worst Visual Variables to display nominal data?
5. You have two cluster Algorithms and want to compare them. They do not necessarily have the same number of clusters, What Visualization do you use?
6. You have 8 discriminants for heart attack risk. You want to find highly correlated variables. What Visualization technique do you use to do this?
 - b) Does this scale with the number of discriminants?
7. Given these sets:
 - pet
 - pet, dog
 - mammal, pet

- mammal, pet, dog

Draw the UpSet Diagramm.

8. I don't remember this question.
9. This Question was similar to the test exam Question for PCA.
10. Three Questions for LDA/PCA. One of them was? Does LDA assume the full rank of the Covariance Matrix? I don't remember the rest.
11.
 - a) What's the difference between regression/classification and ranking in ML?
 - b) What's the intuition for Graph based ranking techniques and which computational Problem do these need to solve?